Bayesian Methods

1 Book: Bayesian Analysis with Python 3rd ed.

Authors: Osvaldo Martin, Christopher Fonnesbeck, Thomas Wiecki, 323 pages. Read up to page: 50

1.1 Thinking Probabilistically

Two main methods in data analysis:

- Exploratory Data Analysis (EDA): numerical summaries: mean mode, standard deviation, etc
- Inferential statistics: making statements about current data: understand a phenomenon or making predictions about the future, or choose between competing explanations of the same set of observations.

Interpret data in the context of models, mental and formal ones.

Models are simplified descriptions of a given system or process that we are interested in [31].

Bayesian modeling process:

- given data and some assumptions, we design a model combining probability distributions.
- We use Bayes'theorem to add data to the model . We conditon the model on our data.
- We evaluate the model

1.1.1 A Probability Primer

Events: subsets of the sample space. Probability: measure of how likely those events are.

$$P(A)=\frac{\text{number of desired outcomes}}{\text{total number of outcomes in the sample space}}$$

The probability of the entire sample space is always equal to 1. So, if the probability of an event increases, the probability of some other event must decrease.

Disjoint event cannot happen at the same time

Axioms of probability:

- the probability of an event is a non negative real number
- $P(S)=P(Sample\ space)=1$
- If A_1, A_2, ... are disjoint events, the probability that any of them will happen is $P(A_1, A_2, ...)=P(A_1)+P(A_2)+...$

1.1.2 Random variables

A random variable is a function that maps the sample space into real numbers. Example each face of a dice is mapped to a number: 1,2,3,4,5,6. Capital letters, like X indicate the variable, lowercase letters indicate the outcome. Example $P(X=3)$: the probability that the outcome is 3. $P(X\leq 3)$: probability that the outcome is less or equal to 3.

The mapping between the sample space and the random variable is deterministic, but the events are random.

The faces of the dice map to variable values, but every time I ask for a new value (toss the dice), I get a different value.

Random variables can be discrete and continuous.

1.1.3 Discrete random variables and their distributions

We are interested in knowing the list of probabilities of all events from the sample space: the probability distribution.

Example: circles. Geometric definition: points on a plane that are equidistant from another point called the center. We can provide a mathematical expression for the definition: $x^2+y^2=r^2$. Given the parameter r , the circle is completely defined.

A given radius r defines a family of circles. From the circle, we can compute properties like the perimeter: $2\pi r$.

For Beta Binomial distribution:

$$P(X=x)=pmf(x)=\binom{n}{x}B\frac{(x+\alpha, n-x+\beta)}{B(\alpha, \beta)}$$

PMF: probability mass function.

1.1.4 Continuous random variables and their distributions

Normal distribution= Gaussian distribution. Probability density function:

$$pdf(x)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

Probability density function (PDF) versus Probability mass function (PMF).

- The output of a PMF is the probability of that given discrete value of the random variable.
- In PDF, each point has zero probability, but an associated probability density So, in a continuous variable, we can calculte the probability that the random variable falls in a given range, not an exact number.

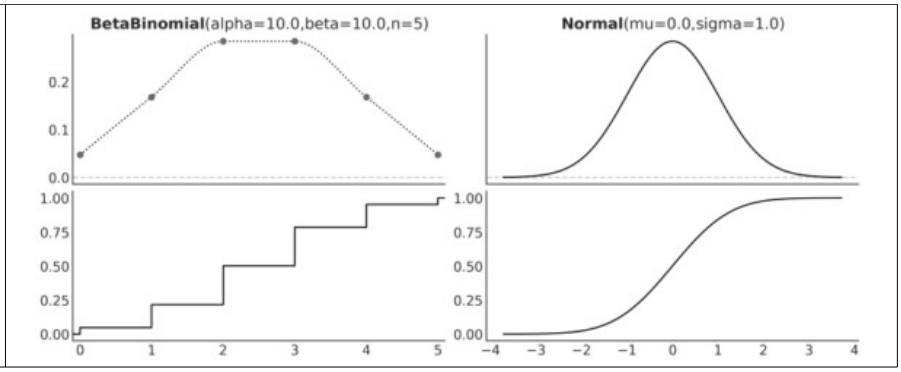
$$P(a<X<b)=\int_a^b pdf(x)\,dx$$

1.1.5 The Cumulative distribution function

Cumulative distribution function: $F_X(x)=P(X\leq x)$

What is the probability that the random variable is lower or equal to x ?

$F_X(x)$ is a smooth function for continuous variable, and it jumps for discrete variables.



1.1.6 Conditional probability

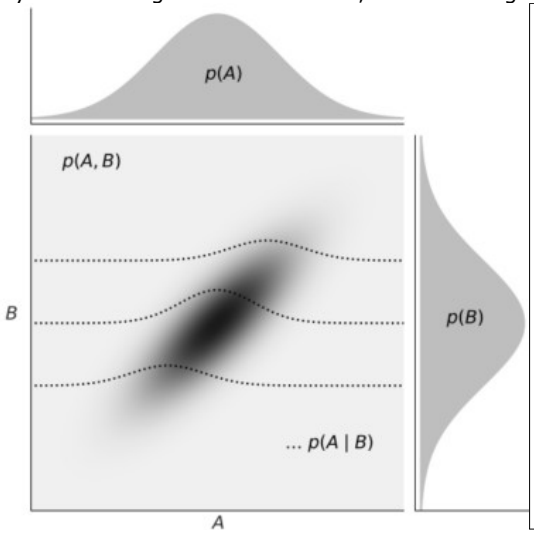
$$P(A|B)=\frac{P(A,B)}{P(B)}$$: the probability that A occurs, given that B has occurred.

$P(A,B)$: the probability that both the events A and B occur.

In case knowing B does not give information about A , then $P(A|B)=P(A)$

Examples: $P(die=3)=\frac{1}{6}$ $P(die=3|\text{die}\in\{1,3,5\})=\frac{\frac{1}{6}}{\frac{1}{2}}=(\frac{1}{3})$

By conditioning on observed data, we are changing the sample space.



$P(A)$ and $P(B)$ are called marginals.

1.1.7 Expected values

In case X is a discrete random variable $E(X)=\sum_x xP(X=x)$. This is the mean or average value of X .

The variance is how much the distribution is spread. The standard deviation is often more useful, which is the square root of the variance. The standard deviation is expressed in the same unit as the random variable.

1.1.8 Bayes' theorem

$$P(A|B)=\frac{P(B|A)P(A)}{P(B)}$$

Proof. From conditional probability:

$$P(A|B)=\frac{P(A,B)}{P(B)} \Rightarrow P(A,B)=P(A|B)P(B)$$

$$P(B|A)=\frac{P(A,B)}{P(A)} \Rightarrow P(A,B)=P(B|A)P(A)$$

$$P(A|B)P(B)=P(B|A)P(A) \Rightarrow P(A|B)=\frac{P(B|A)P(A)}{P(B)}$$

$P(A|B)=\frac{P(B|A)P(A)}{P(B)}$, where: **A**: hypothesis and **B**: data. The probability of **A** given the data **B**.

- $P(A|B)$: posterior. The result of Bayesian analysis, What we now know about the hypothesis once we have seen the

data and the model. The posterior is an updated prior **A** in light of new data **B**.

- $P(B|A)$: likelihood. The expression of the plausibility of data **B** given the parameter **A**
- $P(A)$: prior: what we knew about **A** before seeing **B**
- $P(B)$: marginal likelihood. Called evidence. $P(B)=\int_A P(B|A)\,P(A)\,dA$

The posterior from one analysis can be used as the prior for a new analysis.
The Bayesian Ethos: probabilities are used to measure the uncertainty we have about parameters, and Bayes'theorem is a mechanism to update those probabilities in light of new data.

1.1.9 Single parameter inference

1.1.9.1 Tossing a coin

We assume we have already tossed a coin several times and we have a record of the number of observed heads.
Bias:

- a coin with bias 1 will always land heads
- a coin with bias 0 will always land tails
- a coin with bias 0.5 will land heads an tails half of the time.

θ : bias
Y : number of heads for several tosses.

$$P(\theta|Y)=\frac{P(Y|\theta)P(\theta)}{P(Y)}$$

Choosing the likelihood $P(Y|\theta)=\theta$
Choosing the prior: $P(\theta)$